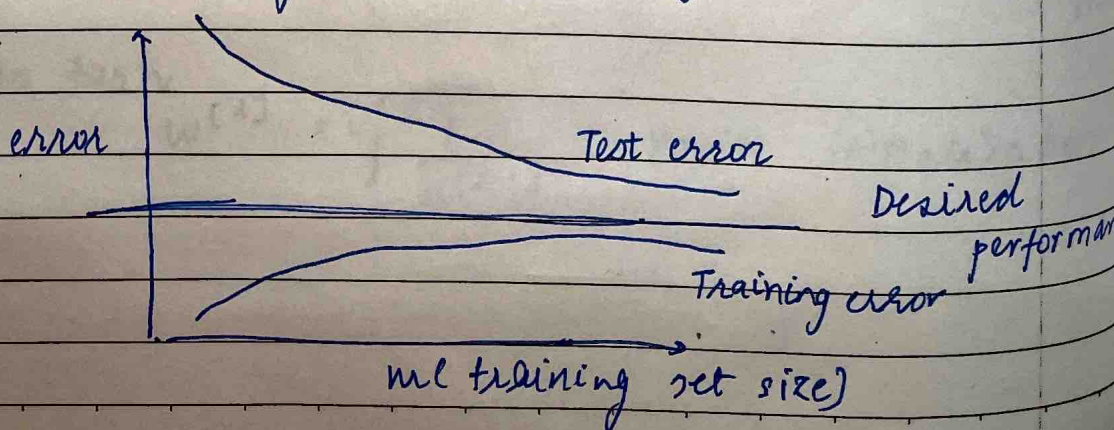


Lecture 13 - Debugging ML Models and Error Analysis

- Advice on getting learning algorithms to work.
- Most of today's material is not very ~~mathematic~~ mathematical, but it's also some of the hardest material in this class to understand.
- Art vs science
- Some of what I'll say is not good advice for doing novel machine learning research.
- Key ideas:
 - 1) Diagnostics for debugging algorithms.
 - 2) Error analyses and ablation analysis
 - 3) How to get started on a machine learning problem.

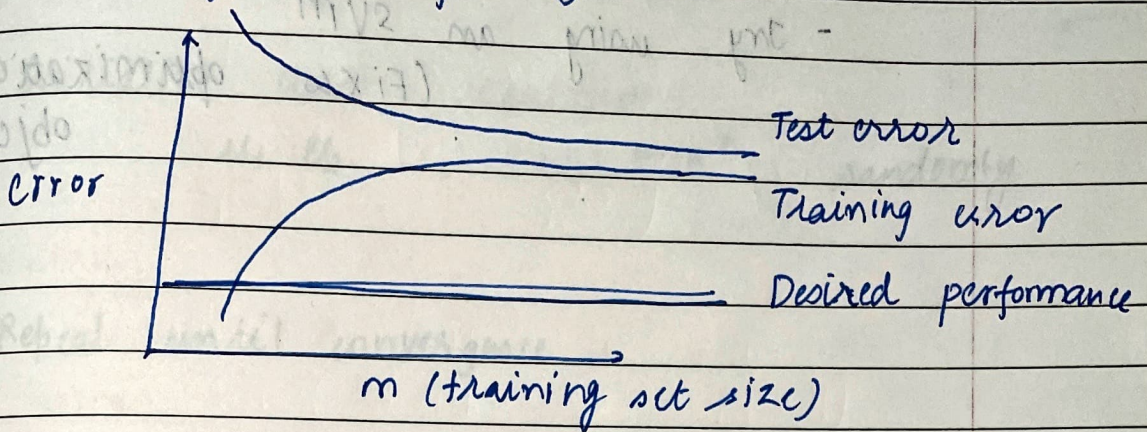
Typical learning curve for high variance:



Test error still decreasing as m increases.
Suggests larger training set will help.

- Large gap between training & test error.

Typical learning curve for high bias



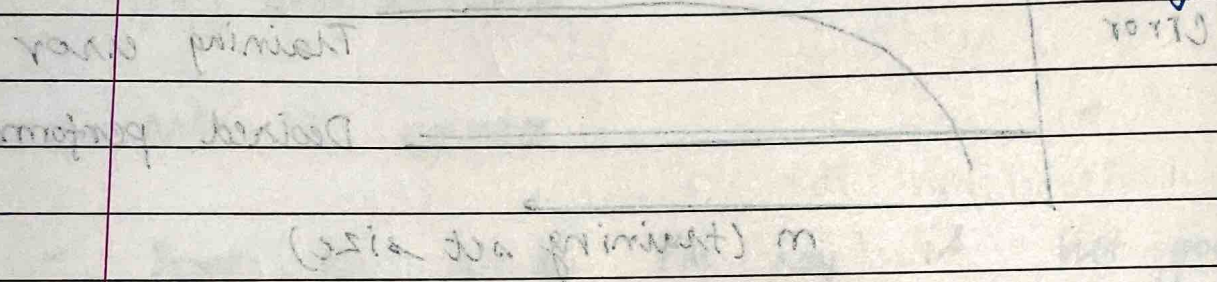
- Even training error is unacceptably high.
- Small gap between training & test error.

Bayesian logistic regression, implemented with gradient descent.

Fixes to try:

- Try getting more training examples (Fixes high variance)
- Try a smaller set of features (Fixes high variance)
- Try a larger set of features (Fixes high bias)
- Try email header features (Fixes high bias)

- Run gradient descent for more iterations (Fixes optimization algorithm)
- Try Newton's method (Fixes optimization algorithm)
- Use a different value of λ (Fixes optimization objective)
- Try using an SVM (Fixes optimization objective)



- Even training error is decreasing but test error is increasing
- This is a sign of overfitting

How to avoid overfitting? -

- try getting more training examples (Fixes high variance)
- try a smaller set of features (Fixes high variance)
- try a larger set of features (Fixes high bias)
- try small number of hidden units (Fixes high bias)